

Published-Receiver Baselines and Perceptual Evaluation for Self-Calibrated Interference Cancellation

Experimental validation note
19 September 2026

Abstract—This note documents a paired receiver comparison that distinguishes published algorithms from internal calibration ablations. A fixed SwinJSCC representation is decoded directly, with a CDDM receiver port, with the original ICDM receiver, or with blind alternating calibration. Pixel error, learned perceptual distance, and multiscale structural similarity are evaluated on the same reconstructions. The study uses development data and is not an independent confirmation or a reproduction of the complete systems’ published performance numbers.

I. WHY THESE BASELINES

The scientific question concerns inference with a fixed transmitter, not a comparison between differently trained image codecs. SwinJSCC [1] therefore supplies the shared codec and a direct-decoding reference. CDDM [2] tests whether a desired-signal prior and Gaussian denoising suffice. ICDM [3] is the closest published receiver: it separately models the desired signal and structured interference. Global-energy and fixed-block-energy variants remain internal ablations; they are not counted as separate external methods.

The CDDM comparison requires an explicit qualification. The original system trains a denoiser and subsequently adapts its decoder. Here, its deterministic denoising update is ported to the same pretrained DiT prior and frozen decoder used by the interference-cancellation receivers. ICDM’s own comparison also uses a DiT-based CDDM, but with a different sampling budget. Our port uses the prior’s discrete training schedule and at most 40 uniformly respaced noise predictions. It is denoted CDDM-RX throughout, rather than presented as a complete reproduction of the original three-stage system.

Two information conditions are retained for CDDM-RX: thermal-noise power only (N_0), and true global noise-plus-interference power (oracle). ICDM retains its published true-global-SINR amplitude and guidance lookup. SC-ICDM A4 receives neither true SINR nor local interference powers. Thus, the oracle rows expose their additional information; they are not described as blind competitors.

II. PAIRED PROTOCOL AND METRICS

The validation set consists of 64 sorted CelebA/CIFAR-10 image pairs, each evaluated with two fixed channel seeds, six SINRs (−7, −4, 0, 4, 7, 10 dB), and stationary, alternating, and burst interference. Noise SNR is 20 dB. Blocks contain 64 complex symbols. All five receiver conditions see identical received samples. The codec, desired-signal prior, and interference prior are frozen. The desired images are 128×128 RGB;

the encoder emits 1024 complex symbols, giving a channel-use ratio of 1/48 per RGB value. The stored codec was trained using image MSE, not the LPIPS training objective of the ICDM paper. These differences preclude copying numerical claims from that paper.

All images are clipped to $[0, 1]$ before evaluation, without resizing or quantization. For image i with D RGB values,

$$e_i = D^{-1} \|\hat{\mathbf{s}}_i - \mathbf{s}_i\|^2, \quad \text{PSNR}_i = -10 \log_{10} e_i. \quad (1)$$

We average e_i and PSNR_i separately. Taking the logarithm of the dataset-average MSE would produce a different PSNR summary. These two metrics describe the same pixel-error family rather than two independent advantages.

LPIPS uses the official VGG-based version 0.1 with images explicitly mapped to $[-1, 1]$; smaller values indicate better perceptual fidelity. MS-SSIM uses four scales at this image resolution, an 11×11 Gaussian window with $\sigma = 1.5$, and the normalized first four published scale weights. Each scale enters the product once; larger values indicate higher structural similarity. The legacy local implementation repeated the last-scale factor, so all methods are re-evaluated with a separately tested implementation. The five-scale implementation agrees numerically with `pytorch-msssim`; the four-scale implementation also passes identity, batch-separation, and analytic constant-image tests. Four-scale scores must not be compared directly with unspecified five-scale scores from another study.

III. VALIDATION RESULTS

Table I reports the complete validation run. The blind A4 receiver obtains the lowest arithmetic mean image MSE and LPIPS and the highest mean MS-SSIM. CDDM-RX with true global SINR obtains the highest mean per-image PSNR, exceeding A4 by 1.011 dB. This is not a numerical inconsistency: PSNR is computed per image before averaging, whereas the MSE column is the arithmetic mean of the per-image squared errors. A small number of difficult images therefore have substantially greater influence on mean MSE than on mean PSNR.

Under exact pairing, A4 improves on direct SwinJSCC by 1.705 dB PSNR, 0.012383 MSE, 0.0907 LPIPS, and 0.1204 MS-SSIM on average. Relative to the global-SINR ICDM receiver, the corresponding changes are +1.424 dB, −0.009535, −0.0353, and +0.0652. By contrast, A4 has higher

TABLE I
 PAIRED RECEIVER VALIDATION WITH A FIXED SWINJSCC REPRESENTATION (64 IMAGE PAIRS, TWO CHANNEL SEEDS, SIX SINRS, AND THREE INTERFERENCE PROFILES; 2304 MATCHED CONDITIONS PER RECEIVER). ORACLE RECEIVERS KNOW THE TRUE GLOBAL SINR; A4 DOES NOT.

Receiver	PSNR (dB) $\uparrow$	Image MSE $\downarrow$	LPIPS-VGG $\downarrow$	MS-SSIM $\uparrow$
SwinJSCC direct	18.866	0.028305	0.4314	0.7067
CDDM-RX (N_0 only)	19.044	0.027870	0.4283	0.7101
CDDM-RX (global-SINR oracle)	21.583	0.019567	0.3425	0.8126
ICDM (global-SINR oracle)	19.147	0.025458	0.3760	0.7619
SC-ICDM A4 (blind)	20.571	0.015922	0.3407	0.8271

PSNR than global-SINR CDDM-RX in only 26.7% of the matched conditions. Its better overall MSE, LPIPS, and MS-SSIM means are driven by large improvements in the most difficult conditions, rather than small improvements in most cases.

The conditional results locate that benefit. Against global-SINR CDDM-RX, A4 changes PSNR by +2.098 and +0.403 dB at -7 and -4 dB SINR, respectively, while also improving all three remaining metrics. From 0 to 10 dB, CDDM-RX is better in all four metrics. A4 is likewise inferior for stationary interference (-2.446 dB PSNR), where a global variance is an adequate description. For burst interference, A4 improves all four metrics, including +0.352 dB PSNR, -0.011719 MSE, -0.0749 LPIPS, and +0.0980 MS-SSIM. Alternating interference lies between these cases: A4 has 0.940 dB lower PSNR but slightly better mean MSE, LPIPS, and MS-SSIM.

These results support a conditional receiver rather than a universal superiority claim: direct decoding at high SINR, global denoising when the interference is approximately stationary, and local self-calibration for strong nonstationary interference. In particular, A4 loses 1.484 dB PSNR to direct SwinJSCC at 10 dB SINR, so a high-SINR bypass is required by the data.

We report arithmetic means, paired mean differences, and the fraction of matched conditions in which each receiver is better. No interval estimate or dispersion summary is attached to this exploratory validation set. The validation images were used in prior model selection, so no independent-confirmation claim is made. Receiver timings are measured after warm-up in the same interleaved run on an RTX 4090, with batch size four. They include inference, output normalization, and decoding, but exclude encoding and metric evaluation. Another project’s training process was detected on the host during execution; the timing records are therefore diagnostic rather than an exclusive-hardware efficiency comparison.

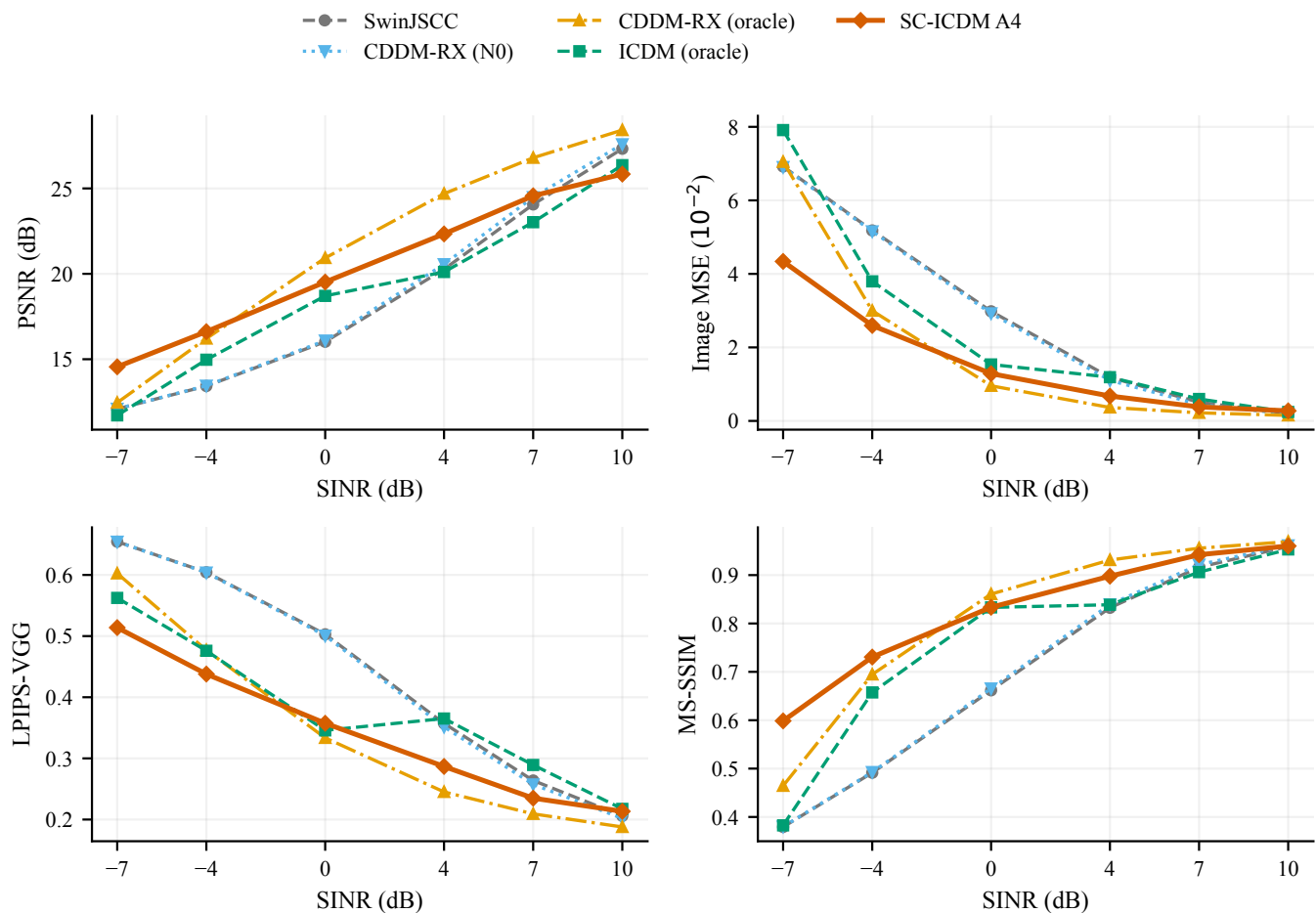
IV. SCOPE OF THE EVIDENCE

The comparison isolates receiver choices under a common representation and records their differing side information. It does not establish superiority over all published complete communication systems. CDDM’s full training pipeline, additional independent receiver families, fading, unknown block boundaries, and a frozen held-out confirmation remain outside this run. The existing 200-pair historical ablations must remain

separate from these 64-pair validation results. Combining them into one larger sample would obscure both the selection history and the evaluation protocol.

REFERENCES

- [1] K. Yang, S. Wang, J. Dai, X. Qin, K. Niu, and P. Zhang, “SwinJSCC: Taming swin transformer for deep joint source-channel coding,” *IEEE Transactions on Cognitive Communications and Networking*, vol. 11, no. 1, pp. 90–104, 2025.
- [2] T. Wu, Z. Chen, D. He, L. Qian, Y. Xu, M. Tao, and W. Zhang, “CDDM: Channel denoising diffusion models for wireless semantic communications,” *IEEE Transactions on Wireless Communications*, vol. 23, no. 9, pp. 11 168–11 183, Sep. 2024.
- [3] T. Wu, Z. Chen, D. He, F. Yang, M. Tao, X. Xu, W. Zhang, and P. Zhang, “ICDM: Interference cancellation diffusion models for wireless semantic communications,” *IEEE Journal on Selected Areas in Communications*, vol. 44, pp. 2528–2543, 2026.



Exploratory paired validation; equal averaging over three interference profiles.

Fig. 1. Four quality metrics versus SINR, equally averaged over the three interference profiles. Each point uses the same 64 image pairs and two channel seeds. CDDM-RX is a shared-prior, frozen-decoder port; oracle methods know the true global SINR. Profile-specific curves and individual records accompany this note. The four plots must not be interpreted as four independent tests.